# A Review on the use of Machine Learning Techniques in Music Recommendation System for Healthcare Management

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***Abstract*— Music is one of the most effective therapies for ailments and pushes patients to heal quickly. By making the patient mentally fit and assisting them in overcoming the ailment, music invokes the patient's thinking emotionally. This survey provides a study of various machine learning techniques and its types utilized in recommendation of music for healthcare management. The machine learning types include supervised, unsupervised techniques and their sub –categories are discussed. The research shows that music therapy improves people's quality of life while progressively diminishing the effects of sickness on the human body Music therapy through a good recommendation system acts as an effective remedy to eliminate the emotional and psychological health issues of individuals. The classification accuracy is considered as a key parameter to define the effectiveness of music recommendation systems. In this survey, machine learning techniques acts as a tool to recommend music for health care management of patients.**

***Keywords—Disease Diagnosis, Healthcare Management, Machine learning, Music Recommendation System, Music Therapy***

## I. Introduction

Music is considered one of the universal languages which help to bring out the emotions of people from all over the world [1]. Music helps to maintain the control of mind and attention, it also helps in strengthening our mind and makes us psychologically fit [1]. It is experimentally proved that listening to music makes our respiratory rate and heart rate normal and helps to lower the stress level [1]. Music therapy experimentally proved in reducing the patient’s stress level, and depression and improve the quality of sleep. Music recommendation systems utilize music stimuli which make increases gamma waves in the brain to improve memory function and the patient’s psychological signs. In consequence, the physiological signs measured on the skin are very useful in analyzing the emotional effects of music [2]. The intelligent background music system is designed and built at the same time using an innovative fusion of IoT and deep learning methods (a branch of machine learning). An intelligently designed background music system has presented outstanding outcomes in data processing for communication systems. [3].

How individuals locate and listen to music is evolving as a result of continuing research in the fields of music information retrieval (MIR) and music emotion recognition (MER). The practice of merging data from MIR and MER has been growing, utilizing different music elements and annotations along with mining data or machine learning techniques [4]. Additionally, Human Activity Recognition uses an inertial sensor for a recommendation system which helps in understanding human behaviors and medical health care systems [5]. The prediction of appropriate music recommendation system for the healthcare is a challenging one because it may vary for person to person according to the mindset of the patient. This study analyzes various works on music recommendation system. This work acts as a tool to helps the future researchers dealing with music recommender systems. The results of the survey provide music recommendation systems in healthcare management using various machine learning techniques.

This research is organized as follows: The taxonomy of the music recommendation system is presented in Section 2. The literature on music recommendation systems is presented in Section 3. The Comparative analysis is presented in Section 4. Section 5 presents the conclusion of this survey paper.

## II. Background

Music has the ability to evoke different emotions of the people. This survey is employed with machine learning techniques, since machine learning techniques are useful for extracting the features effectively and helps to improve the classification accuracy of the music recommendation system [1]. The collection of data is known as a dataset that is used as input for the whole process of recommending music based on the patient’s emotions. The dataset is transferred to the procedure of preprocessing to remove the unwanted spaces, avoid noises and increase the efficiency of the music recommendation system. Figure 1 emotion recognition based on patient emotion [6]. It is necessary to perform the Feature extraction process after the completion of preprocessing to extract the music types such as melody, classical, etc., based on the emotions of the patient. Then the data is transferred to feature selection to decrease the space and time complexity of the discussed model and to enhance the results of classification. A classification process is used to detect emotions such as happiness, sadness, anger, fear, etc. The various classification algorithms used to improve the accuracy rate and recognition rate in the developed music recommendation system. After finding the emotion of the patient the data collection is started to serve the


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recommended music. The recommended music plays only depend on the patient's mindset. To perform the whole process machine learning algorithms are used.

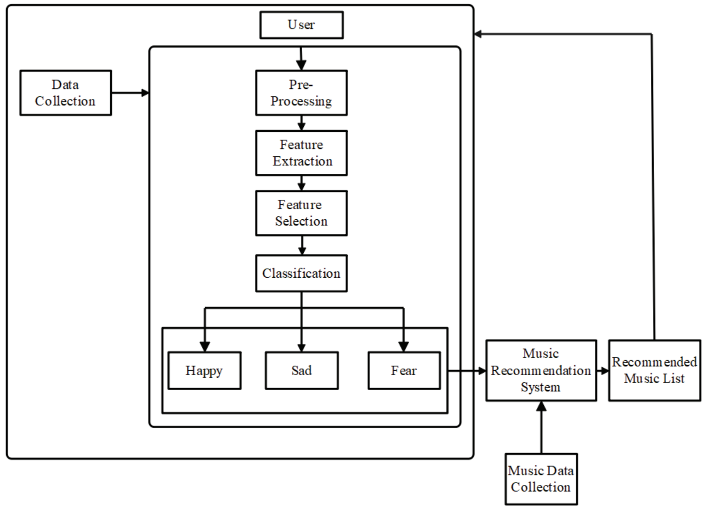


Fig. 1. Emotion Recognition based on Emotion

*A. Taxonomy of music recommendation system:*

Music recommendation is provided for the individuals based on their mindset and likeliness. In general music recommendation is based on five genomes such as rock, electronica, jazz, world music and classical. According to the previous playlist of the user, the nature of the music is identified as recommended for the user. The music recommendation system can be categorized into content based recommendation system and collaborative based recommendation system. Research on recommender systems in the music industry used ML techniques to create representations of songs or artists (embedding or latent factors) from the audio content or textual metadata, including artist biographies or user-generated tags. Then, either directly or indirectly, these latent item factors are applied to content-based systems like nearest neighbor recommendations, integrated into matrix factorization techniques, or utilized to create hybrid systems, most frequently combining content- and collaborative-based approaches [7]. The classification of music recommendation system is represented in fig.2.

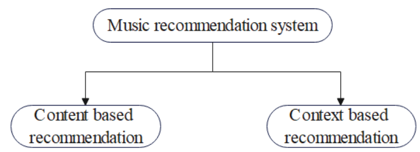


Fig. 2. Types of music recommendation system

*B. Taxonomy based on machine learning techniques:*

Music recommendation system can be provided based on Machine learning (ML) is considered one of the main branches of Artificial Intelligence (AI) which enables software applications to result more accurately at forecasting outcomes and make decisions based only on the user's input. So various ML techniques are used to develop a music recommendation system based on patients' suggestions to enhance their physical as well as mental health [8].

The taxonomy of music recommendation based on Machine Learning algorithms is given in fig 3.

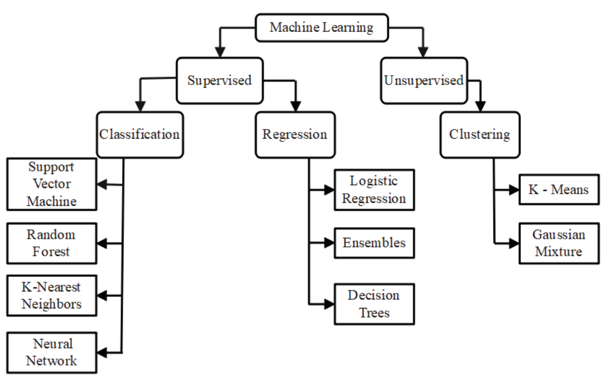


Fig. 3. Taxonomy of Music Recommendation based on Machine Learning

The ML model can estimate the likely emotional state score that a human listener would assign to recently input music pieces. The process of determining which musical characteristics result in demonstrable gains for certain


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applications may be substantially aided and accelerated by machine learning. In the recent years various machine learning techniques have been applied in recommending music for better health-care [1][6][11].

## III. Literature Survey

The ML comprises Supervised and Unsupervised Learning which are discussed below.

### A. *Supervised Learning:*

Supervised Learning [5] is trained using labeled datasets. Supervised learning categorizes into two ways such as classification and regression.

- Classification: The classification separates a group of data into classes based on the assigned values of the object which is used in this study to classify the patient's emotion.
- Regression: The regression is used to forecast the continuous values. The main goal of regression in supervised learning is to design the best curve or best fit line between the data.

The detailed view of machine learning models was given as follows:

#### *1) Support Vector Machine*

Support Vector Machines (SVM) belong to a supervised learning algorithm that is used to classify the emotions such as anger, happiness, sadness, etc. In [3], the SVM removes the overfitting of the user and solves the convex optimization issues. From the analysis, the classified emotions from the SVM are transferred to the music recommendation system. The music will be played based on every transferred music respectively.

#### *2) Random Forest*

The Random Forest (RF) is a supervised learning ensemble classifier used for emotion recognition. In [9-12], the Wavelet Packet Transform (WPT) and Random Forest for emotion recognition and the combination of the hybrid model showed superior performance. The average testing and training time is reduced in the emotion recognition model by using the hybrid RF classifier with WPT. From the above-mentioned statement, the music recommendation based on the recognized emotions is done quickly.

#### *3) K-Nearest Neighbors*

In [8], the K-Nearest Neighbor (KNN) is used for the classification of emotions because the classification accuracy is high when the size of the datasets is small. It is also considered that the accuracy of the KNN classifier emotions substantially increases even the overlapping occurs between emotions. The classification system was tested to identify the emotions of disabled people. Humans with intellectual disabilities and healthy people are tested during the testing phase. It is difficult for human beings to discriminate between subtle emotions. When the size of the dataset is high, recommending music using KNN is difficult.

#### *4) Neural Networks*

In [13], the Artificial Neural Network (ANN) is used to predict the song related to the stress level and also to switch the song based on the continuous monitoring of the patient. Additionally, detailed Music Analytics such as duration and type of music will be performed by the ANN to identify the impacts of music played based on the patient physiological parameters. From the above-mentioned statement, the music recommendation by recognizing the emotions of the patient becomes easier.

#### *5) Logistic Regression*

In [6], the Logistic Regression (LR) classifier is used to classify the patient's emotions such as happiness, sadness, anger, etc., by taking training and testing data. It works on calculating the probability of an event/emotion and classifying the emotion depending on a complex non-linear process. Therefore, in recommending suitable music for patients the LR is more useful.

#### *6) Ensemble*

In [12,14], the Rotation Forest Ensemble (RFE) with various classification algorithms such as SVM, KNN, RF, and DT ANN is used for emotion classification and delivers high performance in classification from the EEG signal data. There are three employment as Multiscale Principal Component Analysis (MSPCA) for noise removal, Tunable Q Wavelet Transform (TQWT) for feature extraction and RFE classifier for classification. The TQWT is coupled with the combination of RF classifier and SVM accurately to deliver high performance in the classification. Then the emotions are moved to a set of collected music data and the music will be playing for each emotion.

#### *7) Decision Tree*

A Decision Tree (DT) uses the conditions for the separation of class and it is a powerful tool used for prediction and classification. In [6], the accuracy and precision of the DT are similar but the recall and f1 score are higher than the compared classifiers when the number of samples is chosen correctly. Among those different classifiers, the DT has wide use in the classification of patients' emotions to recommend music for improving their health.

### B. *Unsupervised Learning:*

Unsupervised Learning [11,15] uses machine learning algorithms to examine unlabeled datasets and Clustering is a branch of Unsupervised Learning.

- Clustering: Clustering is a process of dividing or grouping a number of emotions based on their similarities such as the emotion in a group being more similar to the other emotion in the same group.

#### *1) K- Means Clustering*

The K Means Clustering is an algorithm that comes under the category of unsupervised learning. In [15], K-means clustering is used for classifying the detailed data about diabetes disease and is achieved by categorizing the data into k groups. Each point is given the appropriate cluster centroid using K-means, which computes the k centroid. "Classification Via Clustering" classifier, which is a category of meta classifier that enables the transformation of clustering into classification. To determine the minimum error assignment of class labels, the meta classifier was employed. Whenever, human emotions changes, the allocation of k values also changes dynamically. So, the K-Means clustering process is complex to recommend music.


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*2) Gaussian Mixture Model*

The Gaussian Mixture Model (GMM) is preferred for classifying the different emotions from the EEG signals because the model delivers the classified emotions more accurately than Hybrid Mixture Model (HMM), Vector Quantization (VQ), and Bayesian Network (BN). Truncation and Skew GMM are used in conjunction with the Generalized Mixture Distribution Model (GMDM) to categorize brain signals into various emotions. The GMDM model shows that truncation can be applied to the right or left side, both ends, or both ends of a distribution in cases when the distribution is asymmetric. However, the GMM considers the infinite range and symmetric so there is a chance for misclassification in the patient's emotions. So the probability of using GMM in the music recommendation system is less.

There are various types of recommendation system such as movie recommendation [16], food recommendation, fashion recommendation [17] etc. This section provides a discussion about music recommendation system using various techniques.

Rahman et al. [1] introduced Music Therapy which effectively acts as a treatment for patient's mental health care using the methods of Machine Learning (ML). A comparative study was held on different psychological measures such as Blood Volume Pulse (BVP) Electro Dermal Activity (EDA), Skin Temperature (ST) and Pupil Dilation (PD) while listening to different genres of music. The analysis was done using three classifiers like Neural Networks (NN), Support Vector Machine (SVM) and K Nearest Neighbor (KNN) which were used in evaluating the performance and measures. The patient's health care and the musicogenic epilepsy were improved using the proposed method. The dataset was in biased state, during the time of labeling the stimuli so the prediction capability of the machine learning model becomes poor.

Quasim et al. [2] introduced an emotion-based music recommendation system and classification framework (EMRCF) which help to classify songs with high precision based on a person's emotion. The EMRCF collects information from the history of the playlist and categorizes the song (such as happy, sad, motivational, etc…), whenever the person opened the IoT framework on the device it detected the facial data and played the song accordingly. The EMRCF achieved a high range of accuracy using neural networks in grouping under the classes such as frustrated, pleasant, and relaxed. The user cannot create his/her own soundtracks since EMRCF works automated on the frame of IoT.

Vall et al. [3] suggested Fast Region Convolutional Neural Network (Fast - RCNN) by using a deep learning approach to construct an intelligent music recommendation system. Scale Invariant Feature Transform (SIFT) as the feature extraction algorithm and SVM is the classification algorithm. The entire process was based on deep learning methodology, it provides a better base and recognition for an intelligent music recommendation system. However, tolerance tests and compatibility tests must be done on side of F-RCNN model for transmitting the data and protocol used in the music recommendation system.

Krishna et al. [4] suggested doubly truncated Gaussian distribution model detects the emotions of an immobilized person. This model's primary benefit was that it had an asymmetric distribution, which makes it easier to evoke EEG signal data in symmetrical or asymmetrical form. This method's key contribution was the depiction of the most common emotional states, including anger, surprise, sadness, happiness, disgust, and fear, on a nominal scale of measures. The EEG signal samples' unlimited range was condensed to a set of bounds, and the limitless range was used to detect emotions. However, the doubly truncated Gaussian distribution model finds difficulty in detecting the neutral emotion based on their environment.

Abdel-Basset et al. [5] suggested a Spatio - Temporal Deep Human Activity Recognition (ST-deep HAR) for a supervised dual channel which was used by fusion of inertial sensor data based on Long-Short Term Memory (LSTM). The ST-deep HAR provides two channels, the first one is for learning the sequential features of HAR, and the second one was for the modified residual network. The ST Deep HAR provides the best performance while improving the fine tuning feature selection capabilities. ST Deep HAR helps to achieve the efficiency of Internet of Healthcare Things (IoHT) applications. However, the adversarial generative networks must be improved based on heterogeneous HAR in IoHT applications like device investigation and radar data streams.

Wen [18] have introduced an intelligent music recommendation system utilizing deep learning model and the architecture of IoT. The intelligent music recommendation system was based on feature extraction algorithm which extracts the mid-level features of the scene images. The mid-level features were selected using the Gabor feature algorithm which recognize and recommend music for indoor scenarios. The features were extracted using SIFT feature extraction algorithm and the classification takes place using the SVM classifier. The intelligent music recommendation system processes the data of the complex communication systems. However, deficiencies were found in the test related to tolerance and compatibility during the classification of SVM.

Byrns et.al [19] have introduced a unique music therapy to treat the patients with Alzheimer diseases. The music therapy is provided for the patients through a virtual reality environment which looks like a theatre where the participants perform in the crowd of audience. The participants were instructed to perform memory exercises in the environment of music therapy to decrease the negative emotions of the patients. These type of music therapy proven in optimizing emotion of the disease affected individuals in a virtual reality environment. Moreover, this adaptive music therapy helps to escalate the positive emotions of the patients and enables the memory recall of the patients. However, extension of these music therapy must be provided to more number of people from various region.

Vall et.al [20] have introduced a combination of two feature hybrid music recommendation system which performs collaborative filtering method by integrating information present in the music playlist with a representation of feature vector. The hybrid recommendation system fuses out the irrelevant songs from the playlist based


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on collaborative patterns. The music playlist was taken from two datasets such as Art of mix and 8 tracks and then pre-processed using collaborative filter. The hybrid music recommendation system utilized trained auto-tagger to predict the audio features for songs present in the collection of playlist. However, the hybrid music recommendation has not been optimized to get better training rate and prediction times.

The further chart of detailed literature survey along with methodology, advantages, limitations and performance measures is presented in Table 1.

From table 1, it is shown that accuracy is considered as a significant parameter to compute the effectiveness and efficiency of the music recommendation system.

TABLE I. DETAILED LITERATURE SURVEY ALONG WITH METHODOLOGY, ADVANTAGES, LIMITATIONS AND PERFORMANCE MEASURES

| Author(s) | Methodology | Advantages | Limitations | Performance Measures |
|---|---|---|---|---|
| Tao et al. [21] (2020) | An Attention-based Convolutional Recurrent Neural Network (ACRNN) to classify various features from EEG signals and to enhance the emotion recognition accuracy | Integrating the channel-wise attention into CNN and a self-attention into RNN helps ACRNN to improve the accuracy of EEG emotion recognition in DREAMER and DEAP databases | In this method, the testing and training samples were varied and they were not consecutive to time. Moreover, the samples which are encoded consist of some common information after feature extraction | Accuracy (93.38%) |
| Siddiqui et al [13] (2020) | Body Area Networks (BAN) and Artificial Neural Networks (ANN) for delivering immediate care to patients by automating the procedure of music therapy | To integrate the advanced technologies of ANN and BAN to make an automated delivery of the service of music therapy. | There was difficulty to develop a common music sublayer in the ANN because of the variations in the music recommendations of the patients | Heart rate (60-65 beats per minute), systolic pressure (100-120 mmHg), Diastolic Pressure (80 mmHg) |
| Sarsam et al [15] 2020 | Heuristic mechanism using unsupervised machine learning to identify diabetes using emotions | The emotions of the patients and terminologies related to diabetes were obtained using a heuristic mechanism | At the validation stage, unsupervised learning does not contain labeled data related to diabetes and unsupervised machine learning needs several steps for validation | Accuracy (96.53%), kappa static (95.43%), RMSE (64.66%) |
| Subasi et al[14] (2021) | RFE to achieve emotional EEG signal classification with high accuracy | The TQWT technique was light in weight and the analytical models were simple. It was automated and helped to enhance classification accuracy | In this paper, the classification is only based on positive, neutral, and negative emotions and it doesn't classify particular types of emotions like surprise, anger, or happiness. | Accuracy (93.1%) |
| Kasinathan et al [22] (2019) | Fuzzy Inference Engine to detect the emotional impact of the recommended music/song | By applying the fuzzy logic to the music recommendation, the proposed method has greatly achieved by giving the correct song to improve the user's music listening experience | The method does not fulfill the preference immediately when adaption to the music-playing website | Preference (8.75), focus (5.00) and Relax (8.75) |
| Daly et al [23] (2020) | Brain-Computer Music Interfaces (BCMIs) provide a tool for allowing interaction between ongoing music generation and an individual undergoing music therapy | By the analysis of both EEG/fMRI recordings, the BCMI system successfully adjusts the emotional state by modifying the parameters of musical stimuli | The less number of participants in the BCMI was explained as a potential weakness | Accuracy (68.8%) |
| Hamsa et al [24] 2020 | A Wavelet Packet Transform (WPT) Cochlear Filter Bank and Random Forest (RF) Classifier for emotion recognition | The hybrid WPT-RF algorithm provided better performance in stressful talking and noisy conditions with reduced computational complexity | The hybrid WPT-RF framework has been validated using non-spontaneous input signals. This is due to the less amount of natural speech datasets | Accuracy (98%), Precision (94%), Recall (96%), |
| Ayata, [25] (2018) | Data Fusion Base Emotion Recognition (DFBER) method to improve the performance of music recommendation engines by observing users' emotional states | Performance was enhanced with the use of various types of wearable sensors | A main challenging task in data fusion method signals was directly recognizing the values of valence and arousal. | Accuracy (71.53%) |

## IV. ANALYSIS AND RESULTS

The various methodologies used in music therapy for healthcare management of people in reducing their stress, anxiety and depression and plays a major role in improving their physical and mental health.

A graphical representation of the accuracy values from existing methodologies used in music recommendation is represented in fig.4.

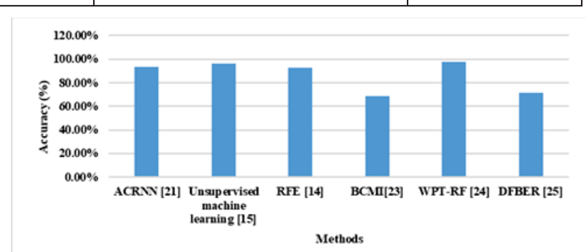


Fig. 4. Accuracy from various music recommendation system


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The analysis shows that the accuracy of the music recommendation is varies from 68%-98%. Still, there is a much improvement is required in this work. The existing researches are affected by some common issues such as irrelevant features, inadequate input data and restricted classification against sub categories of human emotions such as surprise, anger & happiness. Therefore, the future research is directed into the ensemble learning approach for achieving the next level of music recommendation according to the different emotions like surprise, anger & happiness.

The problems found during the development of the music recommendation system from comparative table is discussed below:

1. The repetition of music has some restrictions on the reduction of stress because it extracts long-term memories [10] and possibly negative emotions, and creation of unwanted distractions.
2. The music recommendation system can be added with the personalized music by collecting the past data of the patients so that it helps in fast recovery [9] and well-being of patients
3. Some types of music as well as very loud music might annoy some patients or make them uncomfortable then that type of music could stimulate strong/bad reactions or bring painful memories [6].

From the literature survey, it is revealed that for emotion detection, maximum accuracy is 98% using a Wavelet Packet Transform (WPT) Cochlear Filter Bank and Random Forest (RF) Classifier whereas Brain-Computer Music Interfaces (BCMIs) provide a tool which give minimum accuracy as 68% .

## V. Conclusion

Music recommendation system acts as a tool for the emotion detection of a person, disease impact reduction, and mental healthcare of patients and provides music as a therapy.

In this survey, a music recommendation system is provided for the healthcare management of individuals using machine learning techniques. A related works for music recommendation system are analyzed and taxonomy on types of music recommendation as well as machine learning techniques is discussed. Finally, the problems found from the related works are stated from different techniques. Thus, machine learning (ML) appears to be a crucial new supporting tool for music therapy intervention. ML approaches can give music therapy practitioners highly useful advice for music listening by finding potential predictive elements of therapeutic effectiveness.

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